

Multiple Choice (10 marks)

1. Which body composition method would you use to measure body mineral content?
 - (a) Magnetic Resonance Imaging
 - (b) Dual-energy X-ray absorptiometry
 - (c) Isotope dilution
 - (d) Skinfold Thickness
2. How much dietary protein should athletes consume daily?
 - (a) <0.8 g protein per kg bodyweight per day
 - (b) 0.8-1.2 g protein per kg bodyweight per day
 - (c) 1.2-1.7 g protein per kg bodyweight per day
 - (d) >2.0 g protein per kg bodyweight per day
3. Higher dietary sodium (salt) intake is generally associated with:
 - (a) Decreased calcium excretion in the urine
 - (b) Increased risk of fracture
 - (c) Decreased dietary calcium absorption
 - (d) Negative calcium balance and bone mineral loss
4. Nutrients may pose health risks:
 - (a) When intakes exceed established ULs (Upper Levels)
 - (b) At any level greater than a recommended dietary intake
 - (c) Only when intakes are less than those required for nutritional benefits
 - (d) Under no known conditions of usual intake
5. The effects of iron on certain aspects of socio-emotional development (shyness, orientation/engagement and response to unfamiliar pictures) is due to:
 - (a) The role of iron in myelination
 - (b) The role of iron in neurochemistry
 - (c) The role of iron in dopamine production
 - (d) None of the above

True / False (10 marks)

Answer True or False

6. True or False: The elimination of toxins is the most important function of the gastrointestinal tract right after eating.
7. True or False: The absorption of glucose in the gut is linked to the uptake of sodium ions.
8. True or False: High protein diets can promote weight loss and maintenance through mechanisms like appetite regulation and increased thermic effect.
9. True or False: Type IIb muscle fibers have low oxidative capacity and are not known for high fatigue resistance.
10. True or False: Bariatric surgery can reduce the incidence of type 2 diabetes in obese individuals by 80-90%.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the Atherogenic Lipoprotein Phenotype (ALP) and explain how its lipoprotein abnormalities, including HDL, LDL, and triglycerides, impact cardiovascular health.
12. Discuss the nutritional status of older adults, focusing on the implications of iron levels in their diet. Why might high iron stores pose a greater risk than iron deficiency?

End of Paper